



Electrical and mechanical properties of $\text{Li}_2\text{O}-\text{BaO}-\text{B}_2\text{O}_3$ glass system

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ABSTRACT

Lithium barium borate glasses of composition $x\text{Li}_2\text{O}-5\text{BaO}-(95-x)\text{B}_2\text{O}_3$ ($x = 5, 10, 15$ and 20 mol%) were prepared by a conventional melt quenching method. Structure that was investigated using X-ray diffraction has indicated the amorphous state of the prepared glasses. Increase in density ($2.281-2.495 \text{ g/cm}^3$) and the decrease in molar volume of the glass samples indicated the compactness of glass network upon addition of Li_2O . Measurements of dielectric constant and DC and AC conductivity glass system were carried out at temperatures ranging from 470 to 680 K . Results showed that changes of Li_2O concentration considerably affect the values of dielectric constant and DC and AC conductivities. DC conductivity of glass increases as Li_2O concentration increases. At room temperature and at 4 MHz frequency, longitudinal (L_m), shear (G), bulk (B), Young's (Y) modulus and Poisson's ratio have been obtained as a function of Li_2O content.

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1. Introduction

At first, the applications of glass were within the field of silicate glass [1]. Now, there are some oxide glasses, such as borate, germinate, phosphate and bismuthate. The borate glasses are the best glass formers, which can make glass alone with high chemical durability, good transparency, thermal stability and good rare-earth ion solubility. B_2O_3 glasses do not convert to crystalline state even at low cooling rates [2,3]. Borate glass is the important glass former because it has high bond strength, low cationic size and mostly used as a dielectric material [4].

The addition of alkali oxides to the boron oxide changes some of BO_3 triangles to BO_4 tetrahedral [5–7]. Conversion of triangles to tetrahedral always changes the physical properties, such as increases density and Young's modulus [8,9]. Lithium conducting glasses have large endeavor for utilization in electrochemical devices, phosphors and solar energy converters [10–16]. Due to the most electropositive nature and light weight of Li^+ ion conducting solid electrolytes, it present good prospects in high energy density batteries [17,18].

In this work, we studied the electrical properties (dielectric constant, AC and DC conductivities) of $x\text{Li}_2\text{O}-\text{BaO}-(100-x)\text{B}_2\text{O}_3$ ($x = 5, 10, 15,$

20 mol%) at different temperatures and frequencies. Also, the effect of Li_2O content on mechanical properties of glass system has been studied.

2. Materials and methods

Glass batches were prepared from lithium carbonate Li_2CO_3 , boric acid H_3BO_3 and barium oxide BaO in form $x\text{Li}_2\text{O}-5\text{BaO}-(95-x)\text{B}_2\text{O}_3$ ($x = 5, 10, 15$ and 20 mol.%). Exact mole ratios of the reactants were mixed thoroughly in an agate mortar. The mixtures were initially heated in a ceramic crucible in an electrical muffle furnace at 573 K for 30 min . The furnace was elevated to 1473 K to melt the constituents. The melt was kept in the furnace at the above temperature (1473 K) for about 2 h with steering at several times using quartz rod to ensure the homogeneity of the mixture. The melt was then quenched to room temperature. The obtained samples were annealed by transferring them into another electrical furnace at a temperature of 573 K for 4 h to minimize cracking and thermal stresses of the glasses.

The densities of prepared glassy samples were determined by a simple Archimedes method using an immersing liquid such as toluene. A relative error in density measurements was found as $\pm 1\%$. X-ray diffraction (XRD) investigations for powders of different compositions were carried out using Philips type 1710 chart diffract meter. The patterns of XRD analysis were run in normal 2θ (degree) scan with $\text{CuK}\alpha$ as a target and Ni as a filter ($\lambda = 1.54178 \text{ \AA}$) at 40 kV and 30 mA with scanning speed of $2^\circ/\text{min}$. Electrical properties have been measured in frequency ranges from 10 to 100 kHz and in temperature range from 470 to 680 K by using LCR meter (ChenHwa 1069 LCZ Meter). The

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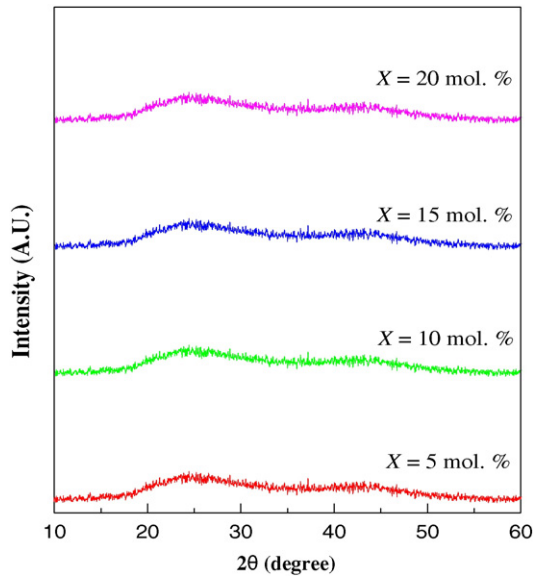


Fig. 1. X-ray diffraction patterns for as-prepared $x\text{Li}_2\text{O}$ – 5BaO – $(95 - x)\text{B}_2\text{O}_3$ glasses.

frequency and temperature were controlled to an accuracy of 0.1% and ± 0.5 K, respectively. Longitudinal ultrasonic velocity was measured in the different glass samples using pulse–echo technique. The accuracy of velocity determination was found to be $\pm 2\%$ for longitudinal velocity while for shear velocity the accuracy was $\pm 4\%$.

3. Results and discussion

XRD of the $x\text{Li}_2\text{O}$ – 5BaO – $(95 - x)\text{B}_2\text{O}_3$ ($x = 5, 10, 15$ and 20 mol%) glasses are shown in Fig. 1. The figure shows no sharp peaks, which means that there is no long range order and it contains only hump. In addition, there is no peak below $2\theta = 20^\circ$, which means that the samples have no medium range order (layer structure). XRD data indicate that the samples have a high degree of the amorphous state [19,20]. Variations of the density and molar volume with the Li_2O content of the respective glasses are depicted in Table 1. The density increases in a systematic way with Li_2O content. Alkali ions resort to occupy interstitial sites within the highly open glass network. Therefore, a large mass exist a little increased volume and the density increases [21].

3.1. Electrical properties

Fig. 2 shows the variation of dielectric constant $\epsilon'(\omega)$ with frequency for $20\text{Li}_2\text{O}$ – 5BaO – $75\text{B}_2\text{O}_3$ glass at different temperature. It is found that, the $\epsilon'(\omega)$ increases with the increasing temperature at lower frequency. The increase in dielectric constant with temperature may be due to the decrease in bond energy. The increase in temperature causes thermal activation and weakening of intermolecular forces which increase the disturbance in orientation vibration which impact the dipolar polarization. At low frequency the ion of lithium (Li^+) can hop from low free

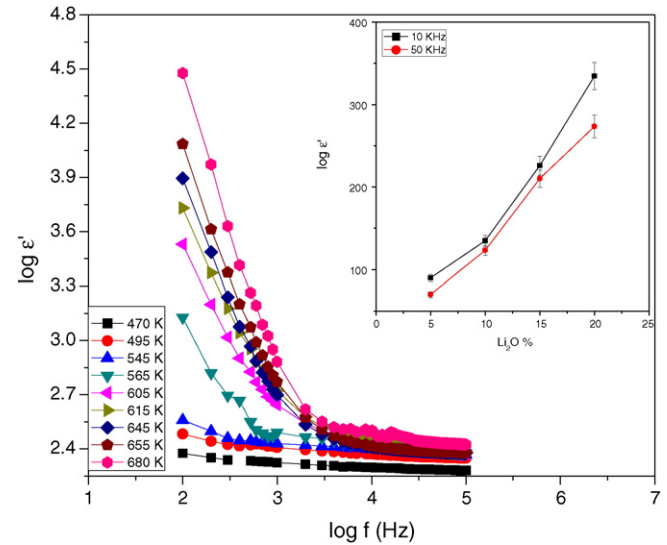


Fig. 2. Variation of dielectric constant $\epsilon'(\omega)$ with frequency for $20\text{Li}_2\text{O}$ – 5BaO – $75\text{B}_2\text{O}_3$ glass at different temperatures. Inset presents variation of dielectric constant $\epsilon'(\omega)$ with Li_2O concentration of glass samples at 680 K.

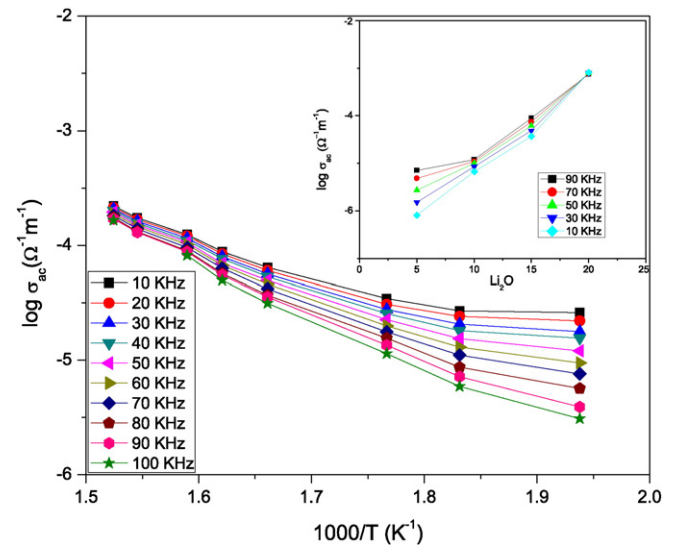


Fig. 3. Variation of AC conductivity $\sigma_{ac}(\omega)$ with inverse temperature for $10\text{Li}_2\text{O}$ – 5BaO – $85\text{B}_2\text{O}_3$ glass at different frequencies. Inset presents variation of AC conductivity $\sigma_{ac}(\omega)$ with Li_2O concentration of glass samples at 645 K.

energy barrier and accumulate at high energy barrier. The net polarization increases due to accumulation increasing causing high dielectric constant at low frequency [22,23]. The conduct of $\epsilon'(\omega)$ with frequency and temperature for the investigated glass systems have demonstrated the same behavior. The variation of dielectric constant with Li_2O content

Table 1

Chemical composition by weight, density, molar volume, longitudinal velocity (V_L), shear velocity (V_S), Poisson's ratio and fractal bond connectivity (F) of glass samples.

Composition (mole fraction)			Density	Molar volume	V_L	V_S	Poisson's	F
Li_2O	BaO	B_2O_3	ρ (g/cm^3)	(cm^3/mol)	(m/s)	(m/s)	ratio	
5	5	90	2.281 ± 0.023	31.4	4420 ± 88	1010 ± 40	0.477	0.217
10	5	85	2.358 ± 0.024	29.6	4800 ± 96	1290 ± 51	0.461	0.319
15	5	80	2.406 ± 0.024	28.2	5140 ± 102	1700 ± 68	0.431	0.532
20	5	75	2.495 ± 0.025	26.3	5810 ± 116	2300 ± 92	0.398	0.760

of glass samples at 680 K is shown in the inset of Fig. 2, which shows that the $\varepsilon''(\omega)$ increases with increasing Li_2O concentration. Increases of dielectric constant may be due to increasing of Li^+ ion which improves the structure of glasses, leading to large dielectric constant [24].

Variation of AC conductivity $\sigma_{ac}(\omega)$ with inverse temperature for $10\text{Li}_2\text{O}-5\text{BaO}-85\text{B}_2\text{O}_3$ glass at different frequencies is shown in Fig. 3. It is clear that, at higher temperature the graph presented linear

tendency and at higher frequency and lower temperature the graph presented plateau. This behavior may be attributed to the existence of polarons in the glass system [19]. Dependence of AC conductivity on temperature has been observed in the same many semiconductor glass systems: lithium germinate [19], vanadium germinate [25], iron bismuthate [26], bismuth vanadate [27] and silver vanadate [28]. The conduct of $\sigma_{ac}(\omega)$ with frequency and temperature for investigated

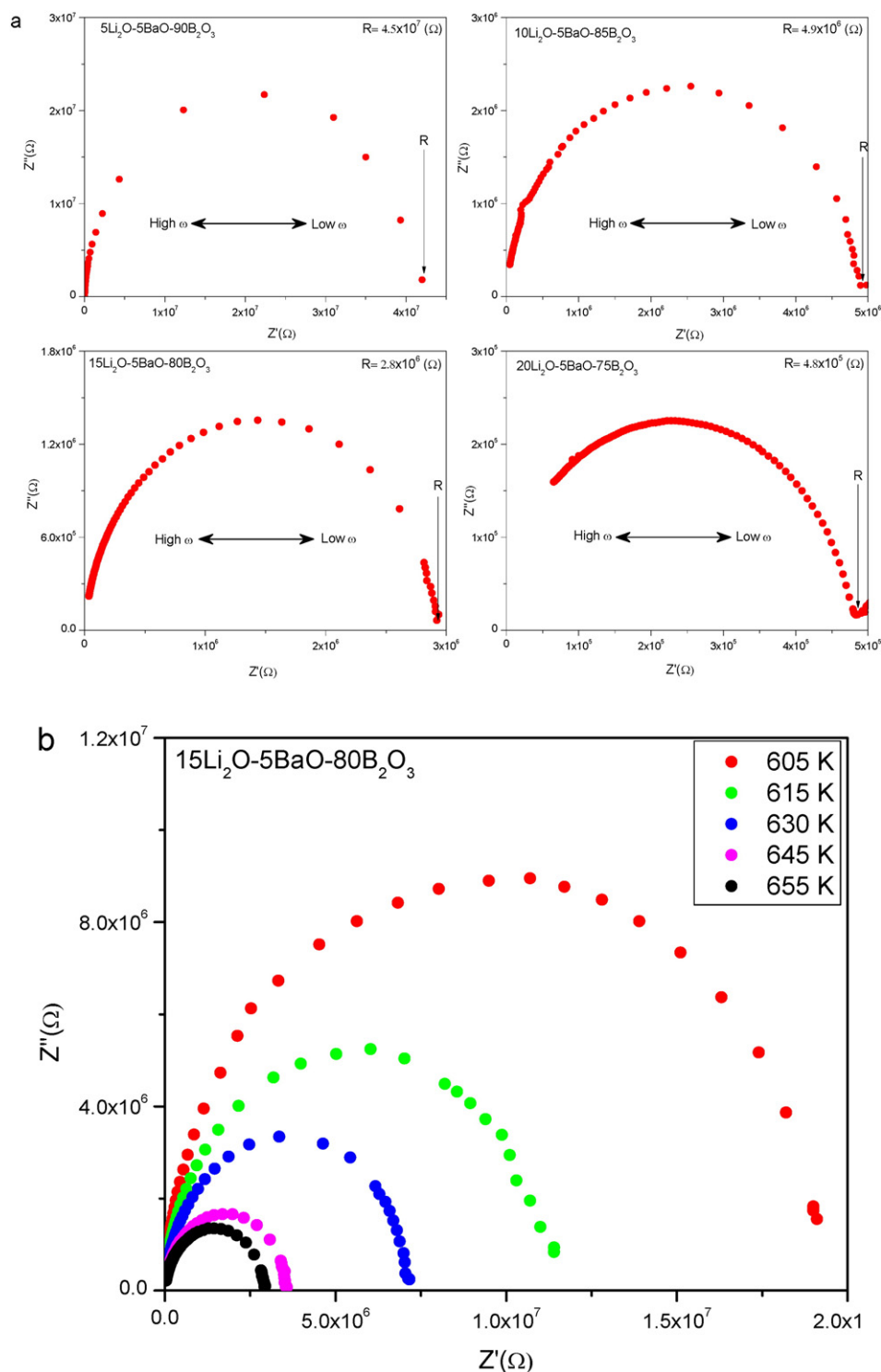


Fig. 4. (a) Impedance spectra for glass samples at 615 K. (b) Impedance spectra for the glass sample $15\text{Li}_2\text{O}-5\text{BaO}-80\text{B}_2\text{O}_3$ at five different temperatures.

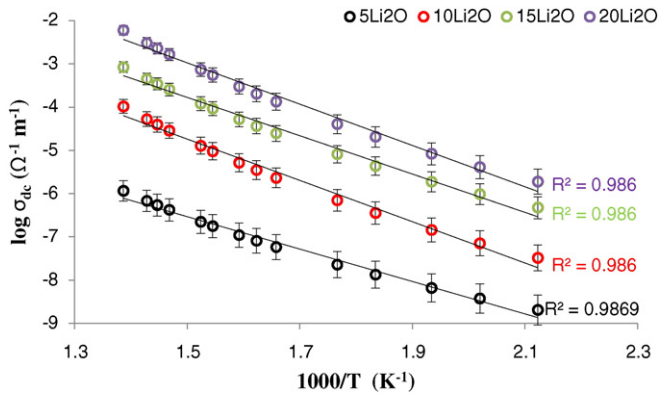


Fig. 5. Variation of DC conductivity of glass systems with inverse temperature.

glass systems have demonstrated the same behavior. The variation of AC conductivity with Li₂O content at 645 K is observed in the inset of Fig. 3. The increase of AC conductivity with increasing Li₂O content may be attributed to that (Li₂O)_x–(BaO)₅–(B₂O₃)_(95 – x) glasses with the substitution of Li₂O for B₂O₃, an increasing number of BO₃ triangles convert into BO₄. As the lithium ions bonded to BO₄ tetrahedra are more mobile, such a substitution causes a large increase in ionic conductivity [5].

To avoid the polarization effects resulting from the use of blocking electrode, the DC conductivity was determined by the AC complex method. Fig. 4 (a) shows the impedance spectra for glass samples at 615 K and Fig. 4 (b) shows the impedance spectra for the glass sample 15Li₂O–BaO–B₂O₃ at five different temperatures. An equivalent circuit that simulates these plots may include a resistor and capacitance in parallel. Each set represents one semicircle, which in turn reflects one conduction process such as bulk, grain boundary conductivity or conductivities through the electrode [23]. The variation of DC conductivity with inverse temperature is given in Fig. 5. It is clear that, the DC conductivity increases with increasing temperature. Fig. 6 shows that, the DC conductivity increases with the Li₂O content increasing. This behavior may be attributed to the increase of charge carriers with Li₂O concentration increasing. On the other hand, the trigonal boron atom will transform into tetrahedral boron atom when the borate glass is modified by alkali oxide (Li⁺). The tetrahedral coordinate boron atom (BO₄) units increase the coherence of the glass network. Each tetrahedral BO₄ has a negative charge and the alkali cations are located in the vicinity of these negative sites, therefore the ionic transport becomes possible [20,29].

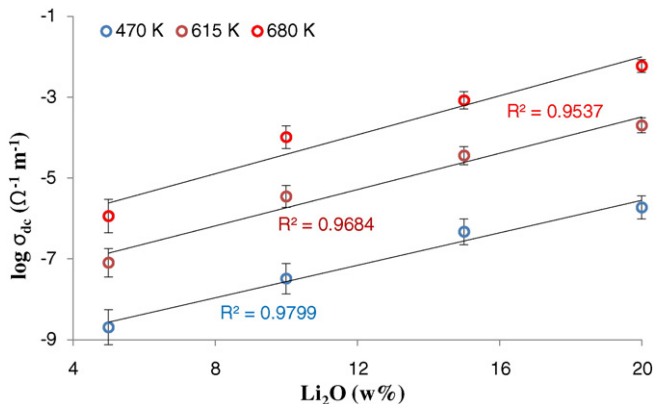


Fig. 6. Variation of DC conductivity with Li₂O content.

3.2. Mechanical properties

Measurement of ultrasonic velocities is a beneficial tool to study the elastic properties of glasses. Velocity of sound provides information on structural modification and the rigidity of glass network. Variation of longitudinal (Lm), shear (G), bulk (B) and Young's (Y) modulus with Li₂O concentration for glass samples are shown in Fig. 7 and calculated by [30]:

$$Lm = \rho V_L^2. \quad (1)$$

$$G = \rho V_S^2. \quad (2)$$

$$B = \rho \frac{(3V_L^2 - 4V_S^2)}{3}. \quad (3)$$

$$Y = \rho \frac{(3V_L^2 - 4V_S^2)}{\frac{V_L^2}{V_S^2} - 1}. \quad (4)$$

It is clear that, the Lm, G, B and Y modulus increase with the increasing Li₂O concentration. Table 1 shows the Poisson's ratio and fractal bond connectivity (F) and calculated by [30]:

$$\text{Poisson's ratio} = \frac{Y}{2G} - 1. \quad (5)$$

$$F = \frac{4G}{B}. \quad (6)$$

The Poisson's ratio decreases with increasing Li₂O content, while the F increases with increasing Li₂O concentration. The composition dependence of Lm, G, B and Y may be attributed to the changes of structural units in the borate network. Pure B₂O₃ glass is essentially constructed by a planar BO₃. The addition of alkali metal oxide (Li₂O) causes the formation of BO₄ and non-bridging oxygen (NBO). The fraction of BO₄ increases monotonically with the increase of Li₂O composition. In this composition range, the influence of the creation of NBO on elastic modulus is much smaller than that of BO₄. The BO₄ units enhance the rigidity of alkali borate glass, because it forms three dimensional covalent bonds in glass network. Hence, Lm, G, B and Y increase with the increase of the Li₂O composition [31].

4. Conclusions

Structure investigations using X-ray have indicated that the glass system is in the amorphous state. The density of prepared glasses increased with increasing (Li₂O) content and the molar volume decreases with increasing (Li₂O) content. The values of the dielectric constant increase gradually with increasing the temperature. The dielectric constant of the present glasses increases with increasing (Li₂O) content, which leads to increase Li⁺ ion density in glasses. This process improves the structure of glasses so as to benefit the ion motions. The dielectric constant (relative permittivity) at a certain temperature decreases with increasing frequency. This leads to a net polarization of ionic medium and hence to a large contribution to the low frequency value of the dielectric constant. At the low temperature range, the AC conductivity is temperature dependent for all composition. The AC conductivity increases with increasing Li₂O content for all frequency ranges (10–100 kHz). The DC conductivity increases with the increasing temperature. The DC conductivity increases with the increasing (Li₂O) content, due to the increase in the number of charge carriers and then decreases the activation energy. The elastic moduli in the ternary lithium barium borate glass increase by increasing (Li₂O) content.

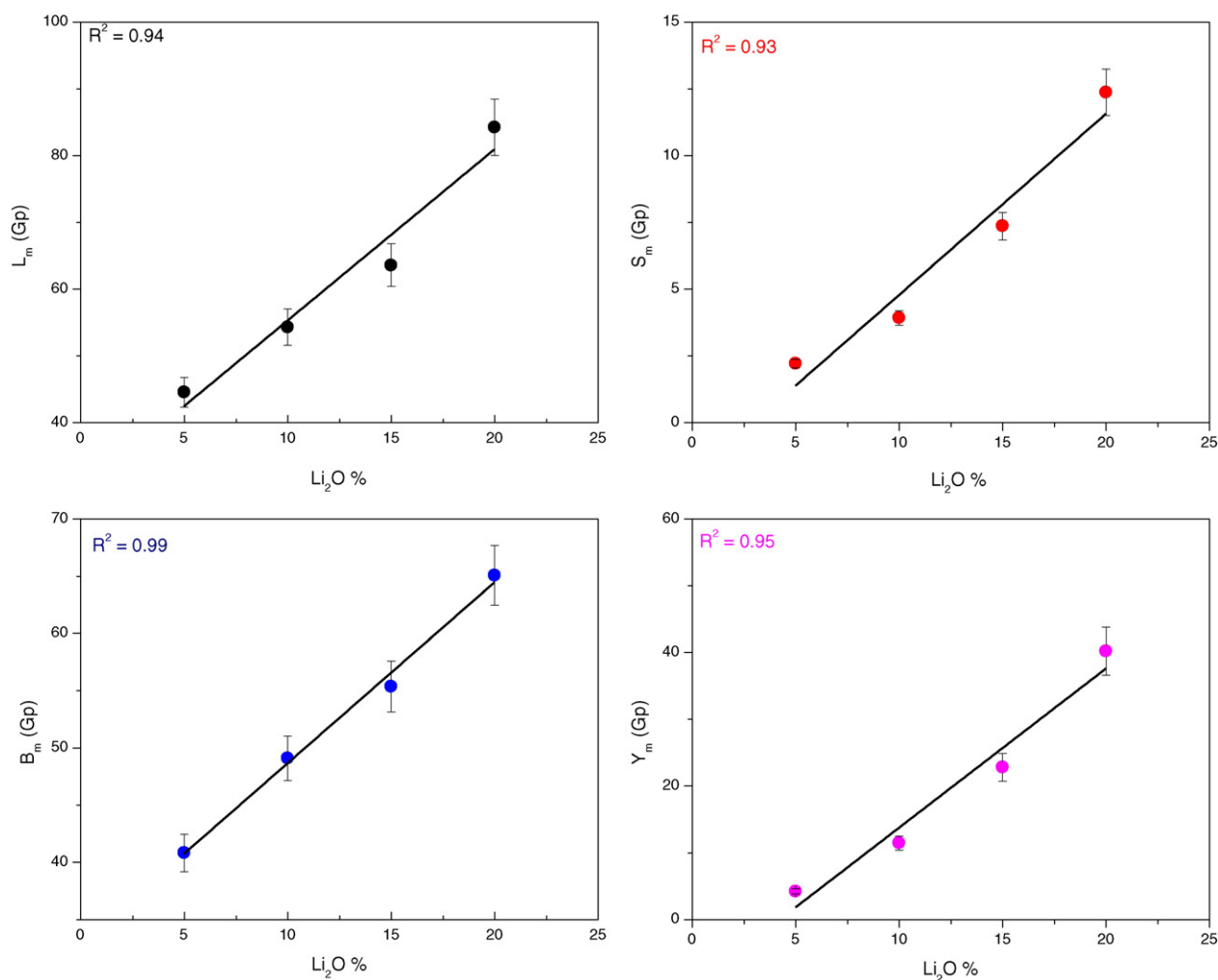


Fig. 7. Variation of the longitudinal modulus (L_m), shear modulus (G), bulk modulus (B) and Young's modulus (Y) as a function of Li_2O content for glass samples.

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